



Light seen physically shift atoms within semiconductor

Researchers at Rice University (Houston, Texas) have observed light physically shifting atoms within a Janus 2D



The finding offers a rare glimpse into how light and matter interact at the atomic level (Credit: <https://interestingengineering.com>)

semiconductor, revealing a phenomenon known as optostriction. When laser light strikes the material, it exerts a tiny mechanical force smaller than a nanometre that distorts its crystal lattice. Using second harmonic generation imaging, the team confirmed this atomic motion, suggesting new possibilities for light-controlled optical chips, sensors, and quantum devices that operate without electrical current.

Your phone's GPS knows more than your location

Researchers at IIT Delhi have discovered that GPS data from Android phones can reveal far more than just



The study was published in ACM Transactions on Sensor Networks

location. Their system, AndroCon, analyses nine GPS signals like Doppler shift and signal strength to detect user activity and even map indoor spaces with under four-metre

error. The finding exposes how apps with GPS access could infer personal details without extra permissions, raising urgent questions about privacy and hidden sensing through everyday technology.

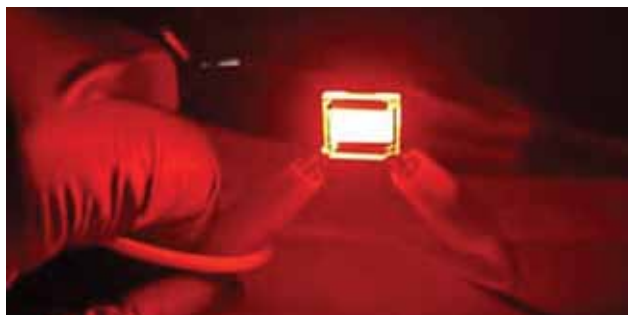
Ultra-thin, glasses-free display could advance 3D visualisation

Researchers at Zhejiang University (Hangzhou, China) have developed an ultra-thin, glasses-free 3D display that measures just 28mm thick, nearly 20 times thinner than conventional systems, while maintaining high resolution

and a 120° viewing angle. Using a directional backlight with freeform optics and micro-prism layers, the display delivers sharp depth and voxels that are six times smaller than those in traditional designs. This breakthrough could advance 3D visualisation in healthcare, education, and entertainment by combining realism with compact form.

Red quantum rod LEDs hit new efficiency record

Researchers at Hong Kong University of Science and Technology have achieved record efficiency in red quantum rod LEDs (QR-LEDs), a step toward brighter, more vivid displays. By refining quantum rod synthesis for uniformity



A sample of the red QR-LED with record-high efficiency developed by the research team (Credit: Hong Kong University of Science and Technology)

and reducing leakage currents through structural redesign, the team achieved a peak external quantum efficiency of 31% and a brightness of 110,000cd/m². The breakthrough could drive next-generation smartphone and TV screens with higher colour purity and energy efficiency.

Wafer-scale memristor for brain-like AI chips

Researchers at the Daegu Gyeongbuk Institute of Science and Technology (South Korea) have developed a wafer-scale process for integrating memristor circuit elements that both store and process data, marking a significant leap toward brain-inspired AI chips. Using a co-design approach with UC Santa Barbara, the team achieved a 95% yield on a 10.16cm (4-inch) wafer and demonstrated scalable 3D stacking. The advance could enable dense, energy-efficient computing architectures that mimic neural processing.

Why perovskite solar cells 'melt'?

Researchers at the University of Colorado Boulder (Colorado) and National Renewable Energy Laboratory, NREL (Colorado) have pinpointed why perovskite solar cells 'melt' under stress: microscopic defects trigger reverse-bias breakdown, causing intense local heating. Using advanced imaging tools, the team discovered that pinholes and thin spots in the film function as hotspots, resulting in short circuits.